

Predicting the prevalence and estimating the economic burden of ischemic heart disease in China

Based on long short-term memory model

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Abstract

Ischemic heart disease (IHD), characterized by myocardial damage from altered coronary circulation, has become a serious threat to the public health of the population. We aimed to predict the future prevalence of IHD and estimate its economic burden in China. Data on the number of patients with IHD in China were derived from the Global Burden of Disease. Long Short-Term Memory network was trained on data from 1990 to 2015, and validated using data to predict the future prevalence of IHD from 2022 to 2030. We used the bottom-up and human capital approach to estimate the direct and indirect economic burden of IHD, respectively. The model demonstrated a strong fit on the training set, and the results of *RMSE* (212,356.03), *MAE* (163,366.30), *MAPE* (0.57%) and *R2* (99.95%) showed that the model fitted well. On the testing set, the results of *RMSE* (364,393.64), *MAE* (321,570.00), *MAPE* (0.54%) and *R2* (97.96%) showed that the model predicted well. From 2022 to 2030, the number of patients with IHD is projected to increase from approximately 66 million to 154 million. From 2022 to 2030, the total economic burden of IHD would be approximately \$220 billion, \$233 billion, \$247 billion, \$265 billion, \$288 billion, \$318 billion, \$359 billion, \$418 billion and \$512 billion, respectively. The prevalence and economic burden with IHD in China are projected to increase substantially in the future. These findings underscore the urgent need for collaborative efforts among the state, society, and the public in the prevention and control of IHD to reduce the economic burden.

Abbreviations: CNY = Chinese yuan, F = forget, GBD = Global Burden of Disease, GNI = gross national income, HR = hazard ratio, I = input, IHD = ischemic heart disease, LSTM = long short-term memory, *MAE* = mean absolute error, *MAPE* = mean absolute percentage error, O = output, PM = particulate matter, *R2* = coefficient of determination, *RMSE* = root mean squared error, RNN = recurrent neural network, USD = U.S. dollar.

Keywords: economic burden, ischemic heart disease, Long Short-Term Memory network, prediction, trend

1. Introduction

Ischemic heart disease (IHD), also known as coronary heart disease, is caused by the narrowing or obstruction of coronary arteries due to atherosclerosis or functional changes (e.g., spasm), resulting in myocardial ischemia, hypoxia, or necrosis.^[1] IHD is the most common cardiovascular disease. There was prevalence rates of IHD of 3222.21 per 100,000 persons in 2021 worldwide.^[2] The number of patients with IHD was 254.3 million in 2021 globally, accounting for more than one-third of people with cardiovascular disease.^[2] In China, the prevalence

of IHD has been rising continuously.^[2] By 2021, the prevalence rate reached 4451.34 per 100,000 persons, far exceeding the global average.^[2] Besides, China had the highest patients with IHD. The number was 63,331,312 in 2021, which was approximately 3.25 times as high as that in 1990 (19,505,463).^[2] Moreover, IHD is a leading cause of death worldwide in 2021.^[3] Globally, the mortality rate of IHD in 2021 was 113.94 per 100,000 persons, and the number of deaths was approximately 9 million.^[2] In China, IHD consistently ranked second among causes of death from 2006 to 2021.^[3] In 2021, the mortality rate of IHD in China was 137.5 per 100,000 persons, and the

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

The authors have no conflicts of interest to disclose.

The datasets generated during and/or analyzed during the current study are publicly available.

Supplemental Digital Content is available for this article.

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How to cite this article: Teng Y, Du H, Chen Y, Chen X, Cao Z, Yu F. Predicting the prevalence and estimating the economic burden of ischemic heart disease in China: Based on long short-term memory model. *Medicine* 2025;104:50(e46022).

Received: 19 June 2025 / Received in final form: 17 August 2025 / Accepted: 14 October 2025

<http://dx.doi.org/10.1097/MD.00000000000046022>

number of deaths was approximately 2.0 million.^[2] As a result, IHD has become a severe threat to public health in China.

The large number of patients with IHD and the chronic nature of the disease impose a heavy economic burden, which is defined as the economic loss caused by the disease. The burden is typically divided into direct economic burden and indirect economic burden.^[4] The direct economic burden is the total costs directly attributable to the control and treatment of disease.^[5] The indirect economic burden is the productivity loss to the society and patient due to morbidity, disability or premature death.^[5] In the United States, the economic burden of IHD is projected to double, from \$10.9 billion in 2010 to \$21.9 billion in 2030.^[6] In the Netherlands, healthcare expenditures for IHD reached \$1.6 billion in 2019 in total, accounting for 1.5% of total healthcare expenditures.^[7] In Turkey, the economic burden of IHD was \$5.9 billion in 2016.^[8] Moreover, the cost of treatment for IHD in China continued to increase from 2012 to 2017.^[9] In 2017, it was \$20.9 billion, accounting for 4.6% of the total cost of treatment for disease.^[9] These figures highlight that IHD is not only a severe public health crisis but also a major impediment to sustainable development. In Australia, a study predicted that there would be 292,249 new incident IHD cases in the Australian working-age population (15–69 years) in 2020 to 2029, with an increased productivity loss of about \$14.80 billion.^[10] Preventing 10% or 20% of new incident IHD cases over 10 years would save about \$1.46 billion or \$2.93 billion.^[10] This study aims to predict the future prevalence of IHD in China and estimate its corresponding economic burden. The results are intended to provide a critical evidence base for developing effective prevention and control strategies.

We employed a Long Short-Term Memory (LSTM) model, a type of recurrent neural network (RNN) designed to model long-range dependencies in sequential data. The model is widely employed in applications such as speech recognition, natural language processing, and time series forecasting, where capturing complex temporal patterns is essential. Besides, previous studies had utilized this method to predict future trends. For example, using LSTM-based prediction models, the morbidity and mortality rates due to breast cancer across 42 low- and middle-income countries exhibited an increasing trend from 1990 to 2019, with projections indicating that this upward trend will continue through to 2050.^[11] Recently, using this method, there were increasing trends in the number of total confirmed cases of COVID-19 in the next 30 days in Pakistan based on

the data from March 11, 2020 to May 31, 2020.^[12] It was also used to predict continued increasing trends in Hepatitis C^[13] and tuberculosis cases in mainland China from January 2018 to December 2020,^[14] as well as the incidence of hemorrhagic fever^[15] and hand, foot, and mouth disease in Fuzhou^[16] from 2010 to 2021. Given this strong precedent, LSTM provides a reliable and powerful methodology for our analysis of IHD trends in China.

Previous studies in China have estimated the burden of IHD mainly through indicators of morbidity, mortality, years lived with disability, years of life lost and disability-adjusted life years.^[17,18] Whereas, most studies estimating the economic burden of IHD have focused on a particular province or certain cities.^[19,20] In this study, we estimated the economic burden of IHD at the national level. We used LSTM model to fit and predict the patients with IHD in China. A bottom-up approach was used to estimate the direct economic burden, and the human capital approach was used to estimate the indirect economic burden of IHD.

2. Materials and methods

2.1. Data

The number of patients with IHD from 1990 to 2021 each year was derived from the Global Burden of Disease (GBD) (<https://vizhub.healthdata.org/gbd-results/>), as shown in Table A in supplementary information. The per capita cost of IHD treatment in 2021 was obtained from the China Health Statistics Yearbook (<https://www.nhc.gov.cn/mohwsbwstjxxzx/tjtjnj/202305/49beded3bd984669bfe9089c6f231cf5.shtml>). The Chinese gross national income (GNI) per capita and the exchange rate in 2021 between U.S. Dollar (USD) and Chinese Yuan (CNY) were obtained from the China Statistical Yearbook 2023 (<https://www.stats.gov.cn/sj/ndsj/2023/indexch.htm>). In this study, we used 1 USD = 6.45 CNY as the currency.

2.2. Long short-term memory model

The LSTM has a similar structure to RNN, but with enhanced memory units.^[16] By introducing gate control mechanisms, LSTM models address the long-term dependency problems and the gradient vanishing and exploding issues present in RNN models. LSTM units consisted of input (I), forget (F), and output

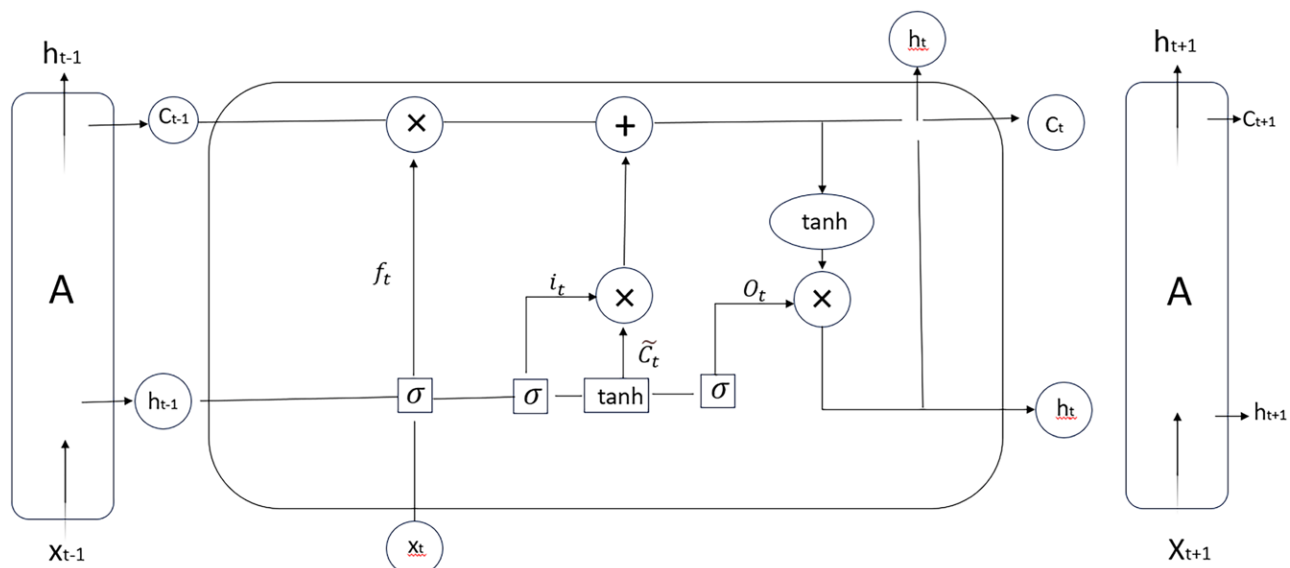


Figure 1. Memory cells of the long-short term memory neural network.

(O) gates. The F gate decided the determines what information to discard, the I gate decides what new information to store in the cell state, and the O gate controls what the next hidden state will be (Fig. 1).

2.3. Statistics

Firstly, we used an LSTM model to fit the number of patients with IHD in China from 1990 to 2015. Root mean squared error (RMSE), mean absolute error (MAE), mean absolute percentage error (MAPE) and coefficient of determination (R^2) were used to evaluate the goodness of fit. Secondly, we used the number of patients with from 2016 to 2021 as the validation set and evaluated the model's performance using RMSE, MAE, MAPE and R^2 . Finally, we predicted the number of patients with IHD from 2022 to 2030. Data analysis was conducted by Python 3.10.11 and Microsoft Excel 2019.

2.4. The economic burden of IHD

The bottom-up approach estimated the direct cost of illness based on the unit cost of providing health care services multiplied by the number of inputs.^[4] The human capital approach measures the loss of production due to morbidity and mortality by estimating the value of future production or income as the present value.^[4]

2.4.1. Direct economic burden evaluation method. We used the cost of IHD treatment per capita in 2021, the patients with IHD in 2021, and the patients with IHD predicted from 2022 to 2030 to calculate the direct economic burden. The formula was as follows^[21]:

$$\begin{aligned} &\text{Direct economic burden of IHD} \\ &= \text{annual cost of treatment per capita for IHD} \\ &\times \text{number of patients with IHD} \end{aligned} \quad (1)$$

2.4.2. Indirect economic burden evaluation method. Indirect economic burden and indirect economic burden per capita of IHD in 2021 were calculated by GNI per capita in 2021, disability-adjusted life years (derived from GBD) and productivity weights for different age groups of IHD in China in 2021. The productivity weights of different age groups were as follows: 0.15 (0–14 years), 0.75 (15–44 years), 0.8 (45–59 years), and 0.10 (60 years or older).^[22] The formulas were as follows^[22]:

$$\begin{aligned} &\text{Indirect economic burden of IHD in 2021} \\ &= \text{GNI per capita in 2021} \times \text{DALYs of IHD in 2021} \\ &\times \text{productivity weights} \end{aligned} \quad (2)$$

$$\begin{aligned} &\text{Indirect economic burden} \\ &\text{of IHD per capita in 2021} = \text{indirect economic} \\ &\text{burden of IHD in 2021} \\ &/ \text{the number of patients with IHD in 2021} \end{aligned} \quad (3)$$

$$\begin{aligned} &\text{Indirect economic burden of IHD} \\ &\text{in future years (2022 – 2030)} \\ &= \text{indirect economic burden} \\ &\text{of IHD per capita in 2021} \\ &\times \text{the number of patients with IHD} \\ &\text{in future years (2022 – 2030)} \end{aligned} \quad (4)$$

2.4.3. Economic burden evaluation method. The economic burden of IHD was calculated as follows:

$$\begin{aligned} &\text{Total economic burden of IHD} \\ &= \text{Direct economic burden} \\ &+ \text{Indirect economic burden} \end{aligned} \quad (5)$$

2.5. Ethics statement

This study was based on data from the GBD Study 2021, which is a publicly available database. All data used were aggregated and fully anonymized, containing no individual identifiable information. Therefore, ethical approval from an institutional review board and the need for informed consent were waived for this study.

3. Results

3.1. LSTM model

The results of RMSE (212,356.03), MAE (163,366.30), MAPE (0.57%) and R^2 (99.95%) showed that the model fitted well. Meanwhile, the results of RMSE (364,393.64), MAE (321,570.00), MAPE (0.54%), and R^2 (97.96%) showed that the model predicted well.

3.2. Prediction results

From 2022 to 2030, the number of patients with IHD was predicted to be 66 million, 70 million, 75 million, 80 million, 87 million, 96 million, 108 million, 126 million and 154 million, respectively (Table 1). From 1990 to 2030, the number of patients with IHD showed upward trend (Fig. 2).

3.3. Results of economic burden of IHD

From 2022 to 2030, the direct economic burden of IHD was projected to be \$150 billion, \$159 billion, \$169 billion, \$181 billion, \$197 billion, \$217 billion, \$245 billion, \$286 billion, and \$349 billion (Table S1, Supplemental Digital Content, <https://links.lww.com/MD/Q896>). The indirect economic burden of IHD was projected to be \$70 billion, \$74 billion, \$78 billion, \$84 billion, \$91 billion, \$101 billion, \$114 billion, \$133 billion, and \$162 billion (Table S2 and Table S3, Supplemental Digital Content, <https://links.lww.com/MD/Q896>). The total economic burden of IHD was respectively \$220 billion, \$233 billion, \$247 billion, \$265 billion, \$288 billion, \$318 billion, \$359 billion, \$418 billion and \$512 billion (Table 2).

4. Discussion

The results of this study showed a continuously increasing trend in the patients with IHD in China from 1990 to 2021, and the number would continue to increase in the future. The patients with IHD in China were expected to exceed 80.0 million by 2025. The reasons for the results are as follows.

Firstly, the aging of the population increase in the elderly population. The prevalence rate of IHD increased with age.^[23] By 2022, China had entered the aged society, with the number of people aged 65 and above amounting to about 210 million, accounting for about 14.9% of the total population.^[24] Over a period of 23 years, the number of elderly people in China had more than doubled. China had a narrowing population pyramid structure and become an aged society in 2021,^[25] so there will be more elderly people in China in the future. Thus, more Chinese people will suffer from IHD in the future. With aging, blood vessels are more prone to sclerosis, and elderly population tends to have multiple underlying diseases simultaneously.^[26] These might

Table 1
Actual, fitted, and predicted the number of patients with IHD from 1990 to 2030

Year	Actual value	Predicted value
1990	19,505,463	19,505,463
1991	20,202,201	20,202,201
1992	20,924,584	21,463,840
1993	21,664,756	21,975,540
1994	22,449,988	22,555,874
1995	23,248,862	23,155,484
1996	24,139,126	23,964,500
1997	25,131,504	25,073,122
1998	26,186,956	25,807,572
1999	27,319,198	26,889,070
2000	28,416,026	28,498,396
2001	29,603,054	29,724,900
2002	30,914,192	30,962,894
2003	32,268,984	32,305,980
2004	33,655,648	33,640,220
2005	34,918,420	35,063,456
2006	36,249,424	36,597,912
2007	37,701,988	37,852,372
2008	39,185,120	39,340,464
2009	40,747,952	40,911,372
2010	42,245,268	42,394,036
2011	43,817,868	43,894,460
2012	45,439,232	45,425,032
2013	47,086,688	47,044,136
2014	48,896,324	48,727,808
2015	50,649,640	50,539,316
2016	52,548,109	52,548,109
2017	54,477,861	54,477,861
2018	56,421,672	56,580,096
2019	58,507,368	58,787,564
2020	60,520,844	61,129,416
2021	63,331,312	63,570,400
2022	-	66,481,224
2023	-	70,200,088
2024	-	74,619,816
2025	-	80,053,224
2026	-	86,914,824
2027	-	95,922,816
2028	-	108,289,200
2029	-	126,281,712
2030	-	154,491,968

IHD = ischemic heart disease.

contribute to the susceptibility of elderly population to IHD. Secondly, the increase in chronic diseases, such as hypertension, diabetes, dyslipidemia and obesity, leads the prevalence of IHD. People with these diseases were at high risk for the occurrence of IHD.^[27] Evidence shows that adverse trends in blood pressure, blood glucose, total cholesterol, weight, and Body Mass Index increased the risk of IHD.^[28,29] From 2002 to 2018, the prevalence of hypertension (18.8%–27.5%), diabetes (2.6%–11.9%), dyslipidemia (18.6%–35.6%), and obesity (7.1%–16.4%) had increased in China.^[30] Hence, the prevention and control of chronic diseases in China is urgent and necessary. Thirdly, unhealthy lifestyles contribute to more IHD cases. Unhealthy lifestyles can directly or indirectly increase the risk of IHD.^[31] The chronic unhealthy diet and physical inactivity lead to excessive fat intake or accumulation in the body and deficiencies in nutrients (dietary fiber, vitamin B, etc), resulting in higher blood pressure, blood glucose and lipid levels.^[32,33] Besides, the nicotine and tar in tobacco, and ethanol in alcohol products could cause excitation of sympathetic nerves, an increase in heart rate, an increase in myocardial oxygen consumption and vasoconstriction or spasm, which could lead to IHD.^[34,35] Meanwhile, urbanization exacerbated unhealthy lifestyles, such as sedentary

indoor, shorter periods of outdoor activities, faster eating patterns, the high intake of high oil, high salt and high calorie foods, smoking, excessive alcohol consumption and too much psychological stress.^[36] Adherence to healthy lifestyles would prevent approximately two-fifths of IHD in less than a decade.^[37] China had been urbanizing on a large scale since 2000. This meant that IHD would definitely become one of the most serious public health problems in China in the future. Finally, the effect of air pollution may cause IHD prevalence. High concentrations of particulate matter (PM) increase the incidence rate of IHD.^[38] PM entered the blood and triggered a vascular inflammatory reaction by inhibiting bone marrow stem cells and endothelial progenitor cells, which led to the incidence and development of IHD.^[39] In a prospective cohort study of 29,141 Chinese adults from Ningbo, we observed an association between long-term exposure to particulate matter and IHD risk. For each 6 $\mu\text{g}/\text{m}^3$ increment in PM_{2.5} concentrations, the hazard ratio (HR) for IHD was 1.21 (95% CI: 1.1–1.3, $P < .001$), representing a 21% increased risk. Similarly, a 11 $\mu\text{g}/\text{m}^3$ increase in PM₁₀ was associated with a 12% higher IHD risk (HR 1.12, 95% CI: 1.0–1.2, $P = .02$) during a maximum follow-up period of 6 years. These findings underscore the potential cardiovascular health impacts of ambient air pollution in urban Chinese populations.^[40]

Hence, the state, society and public should collaborate to promote the prevention and control of IHD in order to advance the Health China 2030 plan and achieve the goal of health for the whole population. We make the following suggestions, organized into 3 areas. Firstly, for the prevention of chronic diseases, the state should improve the capacity of primary services, expand the coverage of screening interventions for high-risk groups of people with IHD, and regularly monitor blood pressure, blood glucose and lipids.^[41] The society should make full use of electronic media or conduct regular health lectures and other ways to popularize knowledge about the prevention and control of IHD or other chronic diseases. The public should have physical examinations actively and pay attention to their health in time as well. Secondly, greater efforts should be made to promote healthy lifestyles. Adherence to healthy lifestyle behaviors is poor in China. Only 2.8% of Chinese followed all healthy lifestyles simultaneously from 2015 to 2019 in China.^[42] The state should improve and implement policies to promote a healthy lifestyles, such as encouraging a reasonable diet, advocating for national fitness, and quitting smoking and controlling alcohol consumption.^[43] Society needs vigorously promote policy measures, laws and regulations that support healthy lifestyles. The public should change their own lifestyles to cultivate a sense of individual management of their own health, such as quitting smoking, controlling alcohol consumption, adopting a diet with less oil, less salt, low fat and high dietary fiber, and exercise for at least half an hour a day.^[43] Finally, for the management of air pollution, the state should adjust the energy structure promptly, vigorously promote cleaner production, improve and implement policies and measures in order to reduce PM levels and achieve carbon peaking and carbon neutrality at an early date.^[43] Through these measures, a favorable environment is created for the national community, so as to prevent and control IHD; the society should strengthen publicity and education on protecting the atmosphere, encourage tree planting and low-carbon travel; the public can choose to walk, ride bicycles or take public transportation instead of using private cars. Using no or fewer fireworks at festivals is also advocated. In addition, the public should avoid traveling during poor air quality or be careful to wear masks when travel is necessary.

The economic burden of IHD in China is substantial. The findings also show that the total economic burden of IHD in China is projected to rise from 2021 to 2030. The direct and the indirect economic burden burdens both show an upward trend from 2019 to 2030. The reasons were as follows.

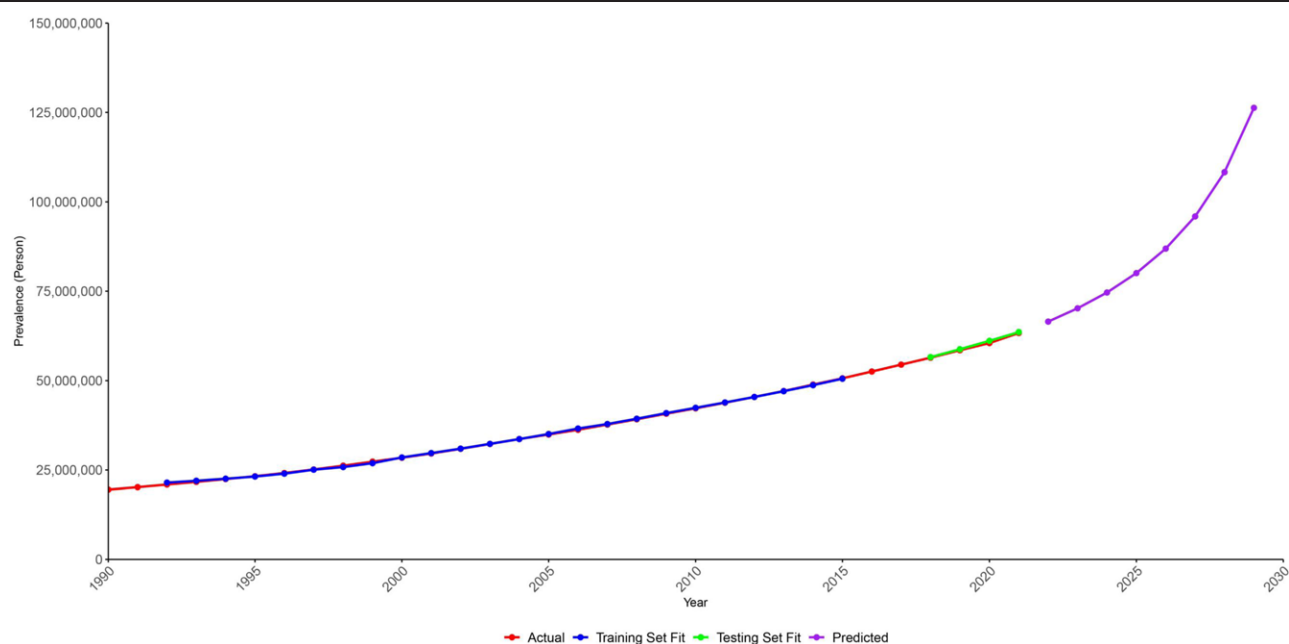


Figure 2. The number of patients with IHD trend from 1990 to 2030. IHD = ischemic heart disease.

Table 2

Total economic burden of IHD from 2021 to 2030 in China (billion).

Year	Direct economic burden	Indirect economic burden	Total economic burden
2021	143	67	210
2022	150	70	220
2023	159	74	233
2024	169	78	247
2025	181	84	265
2026	197	91	288
2027	217	101	318
2028	245	114	359
2029	286	133	418
2030	349	162	512

IHD = ischemic heart disease.

Firstly, the long duration and the large population of people with IHD resulted in a rising direct economic burden. IHD was an incurable chronic disease with a long duration and it had serious complications, such as heart failure, arrhythmia and atrial fibrillation and so on.^[1] It required continuous clinical intervention and inevitably increased the economic burden. Meanwhile, the number of elderly people in China is expected to increase, as life expectancy has risen from 68.6 years in 1990 to 77.9 years in 2020.^[44] Secondly, with the increase in GNI per capita, the disposable income per capita has also increased each year (from \$173.2 in 1990 to \$5337.7 in 2022).^[45] As a result, people with IHD were more likely to access health care services and to choose expensive treatment drugs and treatments, which could contribute to the higher direct economic burden of IHD. Finally, IHD and its complications greatly reduced the quality of life of people with IHD and even led to disability or death. A 2020 questionnaire about 1193 IHD patients in 25 provinces and cities in China showed that more than half of the patients with coronary heart disease in China have poor quality of life status.^[46] About 52.9% of patients still experienced pain or discomfort after treatment; about 35.8% of patients had limited self-care function; about 59.8% of patients had anxiety or depression symptoms.^[46] This would certainly lead to loss of productivity and increase the burden on the state, society and patients. Therefore, the prevention and control of IHD was

urgently needed to protect the health of the population and to preserve the public well-being.

To reduce the economic burden of IHD, we offer the following recommendations. Firstly, the state should formulate and improve medical-related laws and regulations and strengthen medical coverage and protection in order to reduce the cost of treatment for people with IHD. It should also update and publicize dietary guidelines speedily, and actively promote fitness programs, smoking cessation, and alcohol control measures. Secondly, primary care institutions should establish health records for residents and insist on follow-up visits to track of their health status, and carry out nursing interventions. IHD patients should be guided to take lipid-lowering drugs reasonably and their lifestyles should be managed in a standardized way.^[47] Healthy populations should be given reasonable dietary and exercise advice.^[48] Finally, the public should establish the awareness of self-management. For people without IHD, they should cultivate good lifestyle habits to prevent IHD; for people with IHD, they should obey medical advice and actively cooperate with treatment.

4.1. Limitations

There are some limitations of the study. Firstly, according to An Analysis Report of National Health Service Survey in 2018

in China, the proportion of people requiring hospitalization without hospitalization in China was 20.9% in 2018, and the proportion of those who were ill for 2 weeks without treatment was 1.7%. Therefore, the patients with IHD were not equal to the number of consultations or hospitalizations. It would be overestimated to calculate the direct economic burden using the patients with IHD. Secondly, we calculated the indirect economic burden of IHD from 2022 to 2030 using the cost of treatment per capita and GNI per capita in 2021, without considering the effect of inflation. Thus, we might underestimate the indirect economic burden of IHD. Thirdly, the productivity weighting criteria used in the calculation of indirect economic burdens were more than 3 decades old and might have limited applicability to the current social productivity situation in China, and thus the estimated indirect economic burdens might be biased. Adjustment of productivity weights might be one of our future research directions. Finally, in fact, the intangible economic burden was also part of the economic burden. It referred to the psychological pain, discomfort, anxiety and distress of the patient⁵. Based on the current method of estimating the economic burden, we could not accurately estimate the intangible economic burden. In the next step, we would find a feasible method to calculate it.

5. Conclusion

The patients with IHD in China is projected to increase. The economic burden of IHD would become heavier in the future. The state, society and public should collaborate to promote the prevention and control of IHD in order to advance the Health China 2030 plan. LSTM could be applied to predict the patients with IHD in China.

Acknowledgments

We would like to express their gratitude to the Institute for Health Metrics and Evaluation (IHME) for providing access to the Global Burden of Disease (GBD) database through <https://vizhub.healthdata.org/gbd-results/>. The comprehensive data and tools available on this platform have been invaluable in facilitating our research.

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